11 Chemistry — Unit 2 Practice Examination 2026

TEACHER MARKING GUIDE — not for distribution to students

Total 78 marks: Section A 20 × 1 mark (separate answer sheet), Section B 58 marks. Reading time 10 min, writing time 90 min (78 marks × 1.15 min = 90 min). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1}; R = 8.31 J mol^{-1} K^{-1}; c(water) = 4.18 J g^{-1} °C^{-1}). One mark per idea or step; no half marks. Mark units and significant figures once, at the final answer. Where a later part uses an earlier answer, mark it consequentially. This practice paper and guide can be given to students.

Section A — Multiple-choice answer key (20 marks)

Key balance: A ×5, B ×5, C ×5, D ×5; no letter appears more than twice in a row. Each wrong letter points to one slip or misconception.

Q	Key	Working for the key	Distractor diagnosis
1	B	ΔT = q ÷ (m × c) = 12 500 ÷ (400.0 × 4.18) = 7.48 °C; final = 19.0 + 7.48 = 26.5 °C	A: left out c (12 500 ÷ 400.0 = 31.3 °C rise). C: kJ not converted to J (rise of 0.007 °C). D: gave ΔT, not the final temperature. [Easy]
2	D	SLC: 25 °C (298 K) and 100 kPa (Data Book).	A: standard temperature and pressure (STP-type values). B: pressure unit slip. C: confused K with °C. [Easy]
3	A	Cattle digestion (and manure) releases methane.	B, C, D: each adds carbon dioxide, CO_2 (a different greenhouse gas), not methane: confuses the two gases. [Easy]
4	C	KBr 5.00 ÷ 119.0 = 0.0420 M; glucose 10.0 g L^{-1} ÷ 180.0 = 0.0556 M; NaOH 4.00 g L^{-1} ÷ 40.0 = 0.100 M; KCl 0.0600 M	A: compared g L^{-1} values without converting to mol. B: treated 1.00% as the largest number. D: did not convert the other units. [Medium]
5	A	n = 0.200 × 0.0500 + 0.400 × 0.1500 = 0.0700 mol; V = 0.2000 L; c = 0.0700 ÷ 0.2000 = 0.350 M	B: averaged the two concentrations without weighting by volume. C: added the two concentrations. D: gave the amount in mol, not the concentration. [Medium]
6	B	p = 2.00 × 101.3 = 202.6 kPa; n = pV ÷ RT = 202.6 × 5.00 ÷ (8.31 × 298) = 0.409 mol; m = 0.409 × 4.0 = 1.64 g	A: used p = 2.00 (atm not converted to kPa): 0.0162 g. C: gave the amount in mol (0.409), not the mass. D: used T = 25 (°C not converted): 19.5 g. [Medium]
7	D	n = 1.00 ÷ 24.8 = 0.0403 mol; M = 1.13 ÷ 0.0403 = 28.0 g mol^{-1}, which matches N_2 (28.0).	A, B, C: molar masses 20.2, 44.0 and 16.0 g mol^{-1} do not match 28.0; choosing these suggests n was not found from V ÷ 24.8 before M = m ÷ n. [Medium]
8	D	n(C_4H_{10}) = 29.0 ÷ 58.0 = 0.500 mol; n(O_2) = 13/2 × 0.500 = 3.25 mol; V = 3.25 × 24.8 = 80.6 L	A: 1:1 ratio (12.4 L). B: used 13:1, forgetting the 2 in front of C_4H_{10} (161 L). C: ratio inverted, 2/13 (1.91 L). [Medium]
9	C	n(Ca) = 4.01 ÷ 40.1 = 0.100 mol; n(S) = 4.01 ÷ 32.1 = 0.125 mol; 1:1 ratio, so Ca is limiting; S left = (0.125 − 0.100) × 32.1 = 0.80 g	A: thinks the element with the smaller molar mass runs out first. B: compares masses, not amounts in mol. D: right limiting reactant but did not subtract the sulfur used. [Medium]
10	A	n($Ba(OH)_2$) = 0.0450 × 0.02380 = 1.071 × 10^{-3} mol; n(HCl) = 2 × 1.071 × 10^{-3} = 2.142 × 10^{-3} mol; c = 2.142 × 10^{-3} ÷ 0.02500 = 0.0857 M	B: 1:1 ratio. C: halved instead of doubling. D: swapped the two volumes (2 × 0.0450 × 25.00 ÷ 23.80). [Medium]
11	D	Br^- (−1 → 0) loses electrons, so it is oxidised and is the reducing agent; Cl_2 gains electrons (oxidising agent).	A: confuses oxidising and reducing agent. B: thinks the species that changes to an element is reduced. C: reverses electron transfer. [Easy]
12	B	x + 4(−2) = −2, so x = +6 in MnO_4^{2-}	A: +7, the Mn value in MnO_4^- (charge 1−, not 2−). C: +4. D: +2 (reads the ion charge). [Easy]
13	A	$NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O$ (charge: −1 + 4 − 3 = 0 on each side)	B: only one H_2O added to balance O. C: balanced charge ignoring the nitrate's 1−

			charge. D: counted electrons from N = +5 to 0. [Hard]
14	C	Ag^+ is a stronger oxidant than Ni^{2+} (higher in the electrochemical series), so Ni(s) reduces Ag^+(aq).	A, B, D: in each, the metal is a weaker reducing agent than the metal of the ion in solution, so no reaction occurs. [Medium]
15	C	Only tube 3 has both water and oxygen in contact with bare iron.	A: no water. B: no dissolved oxygen. D: paint keeps water and oxygen away from the iron. [Easy]
16	B	Cl ($0 \rightarrow -1$) is reduced, so Cl_2 is the oxidising agent; $2Al \rightarrow 2Al^{3+}$ releases $6e^-$, taken by $3Cl_2$ (each gains $2e^-$).	A: names the reducing agent as the oxidant. C: counts electrons for only one Al atom. D: names the product and counts one Cl_2. [Medium]
17	A	Same number of particles at the same average speed in half the volume, so each part of the wall is hit more often each second, so the pressure is higher.	B: speed only changes with temperature. C: particles do not change size. D: attractions are not the cause of gas pressure. [Medium]
18	D	n: O_2 0.313, CO_2 0.455, N_2 0.429, H_2 0.500 mol; equal mol of gas occupy equal volumes at SLC, so H_2 is largest (12.4 L).	A, B, C: chose by mass, not by amount in mol (B has the largest mass). [Medium]
19	C	Voltage ≈ difference in E°: Mg/Ag 0.80 − (−2.37) = 3.17 V; Zn/Cu 1.10 V; Fe/Pb 0.31 V; Ni/Cu 0.59 V	A: picks the most familiar cell. B, D: metals close together in the series give small voltages. [Medium]
20	B	$q = mc\Delta T$ is the same, so ΔT is inversely proportional to mass: 12 × 100 ÷ 400 = 3.0 °C	A: multiplied by 4. C: thinks the same energy gives the same rise. D: inverted the whole ratio (4 ÷ 12). [Medium]

Section B — Short-answer marking scheme (58 marks)

Accepted answers are in the bold mark allocations; *Reject:* lists common answers that do not score.

Question 1 — Zinc and silver nitrate (10 marks)

a) (1 mark)

1 mark $Ag^+(aq) + e^- \rightarrow Ag(s)$ (states required). *[Easy; L2]*

Reject: Electron on the right-hand side. Ag^{2+}.

b) (1 mark)

1 mark zinc, Zn(s). *[Easy; L1]*

Reject: Zn^{2+} (the product). Ag^+ (the oxidising agent).

c) (3 marks)

1 mark $n(Ag^+) = 0.400 \times 0.0600 = 0.0240$ mol; $n(Zn) = 0.654 \div 65.4 = 0.0100$ mol. **1 mark** 0.0100 mol Zn needs $2 \times 0.0100 = 0.0200$ mol Ag^+, and 0.0240 mol is present, so Zn is limiting (Ag^+ in excess). **1 mark** Ag^+ left $= 0.0240 - 0.0200 = 0.0040$ mol; as $AgNO_3$, $m = 0.0040 \times 169.9 = 0.680$ g (accept 0.0040 mol × 107.9 = 0.432 g of silver ions). *[Medium; L3]*

Consequential: The final mark is consequential on the student's limiting reactant and mol values.

Reject: A conclusion without using the 1:2 mole ratio (0 for the second mark).

d) (3 marks)

1 mark $q = 60.0 \times 4.18 \times (34.0 - 21.0) = 60.0 \times 4.18 \times 13.0 = 3260$ J (3.26 kJ). **1 mark** n(Zn) reacted = 0.0100 mol (zinc is limiting, so all of it reacts). **1 mark** $3.260 \div 0.0100 = 326$ kJ mol^{-1} (3 sig figs, unit required). *[Medium; L2]*

Consequential: Consequential on the limiting reactant in part c). Dividing by $n(Ag^+) = 0.0240$ mol (136 kJ mol^{-1}) loses the second mark.

Reject: ΔT taken as 34.0 °C (853 kJ mol^{-1}). A negative value for "energy released".

e) (2 marks)

1 mark zinc is still limiting, so the same amount (0.0100 mol) reacts and the same energy (about 3.26 kJ) is released. **1 mark** that energy is now absorbed by twice the mass of solution (120 g), so the temperature rise is smaller, about half (≈ 6.5 °C). *[Hard; L4]*

Reject: "Larger, because there is more silver nitrate" (0). "The same, because zinc is limiting" with no reference to the larger mass of solution (max 1).

Question 2 — Hydroponic nutrient solution (8 marks)

a) (1 mark)

1 mark $M(KNO_3) = 39.1 + 14.0 + 3 \times 16.0 = 101.1$ g mol^{-1}; $0.0150 \times 101.1 = 1.52$ g L^{-1} (accept 1.517). *[Easy; L2]*

Reject: Dividing by the molar mass.

b) (1 mark)

1 mark 1.52 g per 1000 mL = 0.152 g per 100 mL = 0.152% (m/v). *[Medium; L2]*

Consequential: Consequential on part a).

Reject: 1.52% (did not divide by 10). 15.2%.

c) (2 marks)

1 mark $n(KNO_3) = 0.0150 \times 20.0 = 0.300$ mol. **1 mark** $m = 0.300 \times 101.1 = 30.3$ g. Accept 1.517 g L^{-1} × 20.0 L = 30.3 g. *[Easy; L2]*

Reject: Dividing by the volume instead of multiplying (7.5×10^{-4} mol).

d) (2 marks)

1 mark mass per tank = 30.3 g (from part c). **1 mark** 500 ÷ 30.3 = 16.5, so 16 full tanks. *[Medium; L3]*

Consequential: Consequential on part c).

Reject: Rounding up to 17 tanks (the 17th tank cannot be completed).

e) (2 marks)

1 mark $n(K^+) = n(KNO_3) = 0.0150 \times 0.250 = 3.75 \times 10^{-3}$ mol. **1 mark** $m(K^+) = 3.75 \times 10^{-3} \times 39.1 = 0.147$ g. *[Hard; L3]*

Reject: Using the molar mass of KNO_3 (0.379 g is the mass of KNO_3, not potassium).

Question 3 — Ammonia in a window cleaner (8 marks)

a) (2 marks)

1 mark $c(NH_3) = 4.5 \div 17.0 = 0.265$ M; n in 10.00 mL = 2.65×10^{-3} mol = n(HCl). **1 mark** $V = 2.65 \times 10^{-3} \div 0.1125 = 0.0235$ L = 23.5 mL (accept 23.4–23.6 mL). *[Medium; L3]*

Reject: Using 4.5 as mol L^{-1}. Answer in L with no unit.

b) (2 marks)

1 mark $n(HCl) = 0.1125 \times 0.01965 = 2.211 \times 10^{-3}$ mol = $n(NH_3)$ (1:1). **1 mark** $c(NH_3) = 2.211 \times 10^{-3} \div 0.01000 = 0.221$ M. *[Medium; L2]*

Reject: Dividing by 19.65 or by 10.00 without converting to L.

c) (2 marks)

1 mark $0.2211 \times 17.0 = 3.76$ g L^{-1} (or compare titres: 19.65 mL measured vs 23.5 mL expected). **1 mark** this is well below the 4.5 g L^{-1} claimed (about 16–17% lower), so the result does not support the claim. *[Medium; L3]*

Consequential: Consequential on part b).

Reject: A conclusion with no numerical comparison (max 1).

d) (2 marks)

1 mark yes, it could: loss of NH_3 lowers the amount of NH_3 in each aliquot, so a smaller titre is needed. **1 mark** this makes the calculated concentration lower than the label, which is the direction observed (a loss of ammonia cannot make a result too high). *[Hard; L4]*

Reject: "No, because it would make the result higher" (wrong direction; 0). "Less accurate" with no direction (max 1).

Question 4 — Biogas from a dairy farm (10 marks)

a) (1 mark)

1 mark water vapour, $H_2O(g)$. *[Easy; L1]*

Reject: Oxygen. Nitrogen. Carbon monoxide (not a product in the equation given).

b) (2 marks)

1 mark n = pV ÷ RT = 850 × 150 ÷ (8.31 × 301) (T in K). **1 mark** = 51.0 mol (accept 50.9–51.0). *[Medium; L2]*

Reject: T = 28.0 (°C not converted; gives 548 mol). Using 24.8 L mol^{-1} (not at SLC).

c) (2 marks)

1 mark $n(CH_4)$ = 0.600 × 51.0 = 30.6 mol. **1 mark** m = 30.6 × 16.0 = 489 g (accept 488–490 g). *[Medium; L2]*

Consequential: Consequential on part b).

Reject: Using 60.0% of the mass of the total gas.

d) (3 marks)

1 mark $n(CO_2)$ from methane = $n(CH_4)$ = 30.6 mol (1:1). **1 mark** $n(CO_2)$ already present = 0.400 × 51.0 = 20.4 mol, so total = 51.0 mol (equal to the total amount of biogas). **1 mark** V = 51.0 × 24.8 = 1.26 × 10^3 L (accept 1260–1265 L). *[Hard; L3]*

Consequential: Consequential on part b).

Reject: Counting only the CO_2 from methane (758 L) or only the CO_2 already present (505 L): max 2. Using 22.4 L mol^{-1}.

e) (2 marks)

1 mark $m(CO_2)$ = 0.400 × 51.0 × 44.0 = 898 g (with $m(CH_4)$ = 489 g from part c). **1 mark** % CH_4 = 489 ÷ (489 + 898) × 100 = 35.3% (accept 35–36%). Also accept a per-mole route: 0.600 × 16.0 = 9.60 g and 0.400 × 44.0 = 17.6 g, so 9.60 ÷ 27.2 × 100 = 35.3%. *[Hard; L3]*

Consequential: Consequential on part b).

Reject: 60.0% (confuses percentage by amount with percentage by mass).

Question 5 — Car airbags (7 marks)

a) (4 marks)

1 mark p = 1.30 × 101.3 = 131.7 kPa and T = 305 K. **1 mark** $n(N_2)$ = pV ÷ RT = 131.7 × 60.0 ÷ (8.31 × 305) = 3.12 mol. **1 mark** $n(NaN_3)$ = ⅔ × 3.12 = 2.08 mol. **1 mark** m = 2.08 × 65.0 = 135 g (accept 134–136 g). *[Hard; L2]*

Consequential: Each later step is consequential on earlier values.

Reject: p used as 1.30 (atm) with R = 8.31 (loses the first mark). Ratio inverted (3/2 gives 304 g).

b) (2 marks)

1 mark in a gas the particles are very far apart, so the volume of the particles themselves is negligible compared with the volume of the gas. **1 mark** so equal amounts (mol) of any gas at the same temperature and pressure occupy the same volume, whatever the mass of the particles (same molar volume). *[Medium; L3]*

Reject: "Argon is a noble gas so it takes up less room." Answers about mass or density only (0).

c) (1 mark)

1 mark V = 3.12 × 24.8 = 77.3 L (accept 77.1–77.5 L). *[Easy; L2]*

Consequential: Consequential on n(N_2) in part a).

Reject: 60.0 L (the volume does not stay the same at different conditions).

Question 6 — Removing iron from a town water supply (8 marks)

a) (1 mark)

1 mark $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$. *[Easy; L2]*

Reject: Electron on the left. Fe(s) as the reactant.

b) (2 marks)

1 mark atoms balanced: $MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$. **1 mark** $5e^-$ on the reactant side, with states: $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$. *[Medium; L2]*

Reject: Balanced with OH^- (not acidic conditions; 0 for the first mark). Electrons on the product side.

c) (1 mark)

1 mark $MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$. *[Medium; L2]*

Consequential: Consequential on the student's half-equations if they are combined correctly.

Reject: Electrons left in the overall equation.

d) (1 mark)

1 mark iron(II) ions, $Fe^{2+}(aq)$. *[Easy; L1]*

Reject: Fe^{3+}. MnO_4^- (the oxidising agent).

e) (3 marks)

1 mark m(Fe^{2+}) = 3.50 × 5000 = 17 500 mg = 17.5 g; n = 17.5 ÷ 55.8 = 0.314 mol. **1 mark** n(MnO_4^-) = 0.314 ÷ 5 = 0.0627 mol. **1 mark** m($KMnO_4$) = 0.0627 × 158.0 = 9.91 g (accept 9.9 g). *[Hard; L2]*

Consequential: Consequential on the 1:5 ratio from the student's overall equation in part c).

Reject: Leaving the mass in mg without converting (gives an answer 1000 times too large). Using a 1:1 ratio (49.6 g).

Question 7 — Comparing galvanic cells (7 marks)

a) (1 mark)

1 mark any one of: it completes the circuit (internal circuit) by allowing ions to move between the half-cells; it keeps each half-cell electrically neutral (balances the charge); it keeps the two solutions apart. *[Easy; L1]*

Reject: "It carries electrons" (electrons flow only in the external wire).

b) (2 marks)

1 mark $Cu + 2Ag^+ \rightarrow Cu^{2+} + 2Ag$ (correct species, balanced for charge). **1 mark** states: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$. *[Medium; L2]*

Reject: $Cu + Ag^+ \rightarrow Cu^{2+} + Ag$ (charge not balanced). Ag(s) reacting with Cu^{2+} (reversed).

c) (1 mark)

1 mark $Al(s) \rightarrow Al^{3+}(aq) + 3e^-$. *[Easy; L2]*

Reject: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ (copper is positive in this cell). Electrons on the left.

d) (2 marks)

1 mark about 1.20 V (1.95 − 0.75; accept 1.18–1.22 V). **1 mark** both cells used the same copper half-cell, and copper was positive in both (aluminium and iron were each the anode), so the difference between the two voltages is the voltage between aluminium and iron; aluminium gives the larger voltage, so it is the stronger reducing agent and would be the anode. *[Hard; L4]*

Reject: Adding the voltages (2.70 V). A prediction from Data Book E° values only, with no use of Zoe's data (max 1).

e) (1 mark)

1 mark Al > Fe > Sn > Cu > Ag (all five in the correct order). Where copper is positive (the cathode), metal X is the stronger reducing agent, and a larger voltage means a stronger one; silver comes below copper because copper is negative (the anode) in that cell. *[Medium; L3]*

Consequential: Consequential on part c) for the position of silver.

Reject: Silver placed above copper or lead (ignores the reversed polarity).

Mark summary

Section	Marks	Parts
A — multiple choice	20	20 × 1 mark; key balance A5 / B5 / C5 / D5
B Q1 — Zinc and silver nitrate	10	a 1, b 1, c 3, d 3, e 2
B Q2 — Hydroponic nutrient solution	8	a 1, b 1, c 2, d 2, e 2
B Q3 — Ammonia in a window cleaner	8	a 2, b 2, c 2, d 2
B Q4 — Biogas from a dairy farm	10	a 1, b 2, c 2, d 3, e 2
B Q5 — Car airbags	7	a 4, b 2, c 1
B Q6 — Removing iron from a town water supply	8	a 1, b 2, c 1, d 1, e 3
B Q7 — Comparing galvanic cells	7	a 1, b 2, c 1, d 2, e 1
Total	78	78 × 1.15 min = 90 min of the 90 min writing time

Study design coverage

Study design dot point	Where practised
Solution concentration (mol L^{-1}, g L^{-1}, % (m/v)) and unit conversions	MC 4; SB 2a, 2c, 2d, 2e, 6e
Volumetric analysis: standard solutions, dilution, acid–base titration	MC 5, 10; SB 2, 3
CO_2, CH_4, H_2O as greenhouse gases	MC 3; SB 4a, 4d (absorbing infrared is practised on the revision worksheet)
Gas pressure and SLC	MC 2, 17; SB 5b
Ideal gas equation pV = nRT with unit conversions	MC 6; SB 4b, 4e, 5a
Stoichiometry of reactions involving gases (mol, mass, volume)	MC 8, 18; SB 4d, 5a, 5c
Molar volume or molar mass of a gas	MC 7, 18 (molar volume from data is practised on the revision worksheet)
Limiting and excess reactants	MC 9; SB 1c, 1d, 1e
Specific heat capacity of water and calorimetry	MC 1, 20; SB 1d
Oxidising and reducing agents; half and overall equations (incl. acidic)	MC 11, 12, 13; SB 1a, 1b, 6a–d
Reactivity series and metal displacement	MC 11, 14, 19; SB 1, 7b–d
Applications of redox: corrosion, simple primary cells	MC 15, 16, 19; SB 7

Paper blueprint

Difficulty and demand by marks.

Measure	Marks	Share
Easy	17	22%
Medium	40	51%
Hard	21	27%
L1 Recall	6	8%
L2 Routine procedure	39	50%
L3 Interpret or explain	27	35%
L4 Evaluate	6	8%